

Liars Are Information:

Receiver-Anchored Channel Estimation for Byzantine-Robust Aggregation in Multi-Agent LLM Systems

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Abstract

When some agents in a multi-agent LLM system lie, the classical defence is to discard them. That defence fails once liars are half the swarm. Prior work showed that a *coherent* liar can be read backwards instead, but it gated this inversion on coherence and then proved that no coherence-based rule survives an adversary coordinating just below the gate. We argue that the informative quantity is *dependence on the truth*. It can be estimated without labels if the estimator is anchored on the one fact a receiver knows: that it is itself honest. We introduce RACE (Receiver-Anchored Channel Estimation). Each honest agent fits an anchored latent-truth model to its own unlabelled history, tempers replicated channels by error-conditioned clone detection, and gives every peer a signed log-odds weight. This weight subsumes the trust/discard/invert gate.

We prove three things. Against a known-channel receiver a lie can only add information, so there is no breakdown point below $f = 1$. Label-free estimators fail at $f \geq 1/2$ by label switching. A receiver anchor provably repairs this, and a near-chance anchor provably cannot. We evaluate on 3,906 replayed worlds (six benchmarks, nine open-weight models, fourteen symbolic attacks, four attacks written by LLMs prompted to deceive) and on a live six-model swarm. At $f = 0.7$, RACE scores 94.9% where the gate-aware adversary drives the prior mechanism to 48.5%. Against real LLM deceivers it scores 89.8%, versus 70.8% for the prior mechanism, 29.4% for majority vote and 85.8% for self-trust, with 0 significant losses in 480 comparisons. Classical crowdsourcing estimators (IWMV, MACE, GLAD, KOS, Dawid-Skene) fall to at most 17.6% in the same regime, which isolates the receiver anchor as the ingredient that matters. Against seven independent liars out of ten, RACE even beats an oracle that knows who lies and removes them (94.8% versus 89.7%). Its label-free channel estimates correlate $r = 0.977$ with the truth. In a live swarm of six small models it lifts every honest agent on MMLU, by 5.6–27.9 points. On binary BoolQ the frozen method fell below the receiver alone, because small models answer yes/no questions with an option bias that a symmetric channel cannot represent. Class-conditional channels for binary questions, which we adopted after this observation and report as such, recover the loss. We report this and the other failure modes (camouflage, sleepers, weak anchors) and an audit that re-derives every number of the prior artifact from repaired data.

1 Introduction

Multi-agent LLM systems increasingly decide by pooling: several agents answer, and each agent combines what it hears. Examples include self-consistency [37], debate [11], judge panels [36] and agent swarms [32]. Pooling fails when some agents are adversarial: compromised by prompt

*Author list withheld for review. This work extends the “Liars Are Information” artifact of D. J. Das [8] and the derivative release of [30]; code, data, results, figures and videos: see the Reproducibility section.

injection [12], fine-tuned to deceive [18], or simply instructed to mislead. The classical answer, inherited from Byzantine fault tolerance [4, 26, 39], is to *discard* outliers. That assumes an honest majority and breaks at $f = 1/2$.

The “Liars Are Information” study [8] made a sharp observation: a liar who reliably says X when the truth is not X is not noise. “This agent says X ” is evidence *against* X . Its mechanism, AIP, inverts a channel when the channel is more *coherent* with other dissenters than honest agents ever are. The same study then defeated AIP. An adversary that knows the published coherence ceiling coordinates just below it, and no rule measurable with respect to per-question coherence can separate it from honest disagreement. Two further failures were documented. On binary questions coherence is degenerate, and AIP scores 0%. And on open-answer mathematics AIP fell below a receiver that simply trusts itself.

This paper. We argue that the premise was right and the statistic was wrong. What makes a liar informative is not coherence but *dependence on the truth*. An agent that never states the truth leaks it by elimination, whether or not it coordinates. An agent whose answers are independent of the truth leaks nothing, however coherent it is. Dependence on the truth is exactly what latent-truth models of annotator reliability estimate [9, 40]. They fail here for a different reason: label-free estimation is identified only up to a relabelling of the truth, and with an adversarial plurality it adopts the liars’ labelling. The fix is the one piece of knowledge a receiver always has: *it is itself honest*.

Contributions.

1. **Race (Receiver-Anchored Channel Estimation).** Each honest receiver fits a one-coin latent-truth model to its own unlabelled history. The fit is anchored on the receiver through initialisation, a prior and a constraint. Replicated channels are tempered by error-conditioned clone detection. The resulting signed, continuous log-odds weight subsumes AIP’s TRUST/DISCARD/INVERT gate (Section 4).
2. **Theory.** (i) Against a receiver that knows the channels, a lie can only add information, and the minimax attack is to be uninformative, so there is no breakdown point below $f = 1$ (Theorem 1). (ii) Label-free estimators must fail at $f \geq 1/2$ by label switching; a receiver anchor restores identifiability, and a near-chance anchor provably cannot (Proposition 3). (iii) AIP’s gate is a quantisation of the Bayes weight, and the coherence impossibility does not bind a truth-dependence estimator (Proposition 5). (iv) With three informative channels the anchored estimator is consistent, and its plug-in decision attains the known-channel accuracy (Theorem 4). (v) We give sample-complexity and clone-detection results that explain the design (Propositions 6 and 7).
3. **Evaluation.** Over 3,906 replayed worlds on six benchmarks with nine models, fourteen symbolic attacks and four prompt-injected LLM attacks, RACE holds at every Byzantine fraction up to 0.9. With real LLM deceivers at $f = 0.7$, RACE scores 89.8% against 70.8% for AIP, 29.4% for majority vote and 85.8% for the receiver alone. Against the gate-aware attacker that defeats AIP, RACE scores 94.9% against 48.5%. Four classical crowdsourcing estimators and both Dawid–Skene variants reach at most 17.6% at $f = 0.7$, where RACE scores 89.5% (Section 7.8). We report where RACE loses as prominently: a camouflage attacker built against it, sleepers that violate stationarity, and weak receivers that cannot anchor.
4. **An information budget for liars.** We compare a known-channel oracle with an oracle that knows who lies and removes them. The gap measures how much information an attack carries, and a negative gap certifies that the attack violates the model. Against independent liars at

$f = 0.7$, RACE scores 94.8%, above the liar-removal oracle’s 89.7%. A rule that only filters liars cannot do that in expectation. Against real LLM deceivers the budget is near zero, and RACE matches the liar-removal oracle in 68 of 72 cells (Section 7.7).

5. **A live multi-agent LLM experiment.** Six small open models from six labs answer 240 questions in honest and covert-saboteur roles, and in a debate condition in which honest agents first see a panel that is half liars (Section 8). This is new inference, evaluated after the method was frozen. On MMLU RACE lifts every honest agent. On binary BoolQ it exposed a misspecification: small models answer with an option bias that a symmetric channel cannot represent. We therefore amended RACE to use class-conditional channels on binary questions (v3.1, Section 4.3), re-ran every study, and report the frozen v3.0 alongside it.
6. **An audit and a repaired, re-derived artifact.** We found that the upstream post-repair recompute was incomplete, found four code defects and a replica-cloning defect, and re-derived every number from the repaired cache (Section 10). All code, data, figures, videos and the claims in this paper are regenerated by one script.

2 Related work

Byzantine-robust aggregation. From the Byzantine generals problem [26] to robust distributed learning, the dominant response to adversarial input is to *remove* it. Krum [4], coordinate-wise median and trimmed mean [39], and Bulyan-style composition [29] are examples. These rules assume an honest majority and break at $f = 1/2$. The original “Liars Are Information” study [8] observed that this ceiling belongs to *discarding*, not to the problem. It proposed Adversarial Informativeness Pooling (AIP), which inverts channels whose coherence exceeds a calibrated honest ceiling. The same study then defeated AIP with an adversary that coordinates just below the ceiling, and proved that no coherence-measurable rule can separate such an adversary from honest disagreement. A derivative release [30] added clean-task calibration, clone caps and causal online windows, with mixed results. RACE keeps the premise (a liar can be read backwards) and replaces the coherence gate with channel estimation.

Learning from unreliable annotators. Dawid and Skene [9] introduced per-annotator confusion matrices fitted by EM [10]. Later work added per-item difficulty [38], spammer-aware models [17], iterative weighted majority voting [27], provably optimal spectral initialisation [40], budget-optimal inference [21], joint learning of a classifier [35], and unsupervised ensemble ranking [31]. Identifiability of such latent-class models holds only up to label permutation [1, 24]. Crowdsourcing assumes a better-than-chance *majority*, which is exactly what fails at $f \geq 1/2$; E11 (Section 7.8) confirms empirically that IWMV, MACE, GLAD, KOS and Dawid–Skene all collapse there. Our contribution is to replace that assumption with a *receiver* anchor (Proposition 3) and to handle replicated channels, which crowdsourcing rarely faces. The anchor resembles semi-supervised crowdsourcing, which seeds EM with gold questions or a trusted annotator, but it needs no gold: each receiver trusts only itself, every honest agent runs its own fit, and no party has to be trusted by the others. The “surprisingly popular” algorithm [34] also extracts truth from a minority. It needs a second elicitation (predicted votes) that an adversary could forge, whereas RACE uses only the answers themselves.

Correlated LLM errors. LLMs make correlated mistakes [22]. Judge panels are worth far fewer independent votes than their size suggests [23, 36, 41]. Open-ended outputs are homogeneous across models [19]. Classical jury theorems degrade under correlated votes [25]. These findings

motivate our error-conditioned clone detection (Proposition 7). We treat correlation as a quantity to estimate from errors, not from raw agreement.

Multi-agent LLM systems under attack. Sampling-and-voting [37] and multi-agent debate [11] improve reasoning when every agent is honest. Adversarial debaters can pull a group toward wrong answers [2]. Deceptive behaviour can persist in LLMs [18, 33], and agents can be subverted by indirect prompt injection [12]. Our live experiment (Section 8) measures the debate side directly. Exposing small honest models to a panel that is half liars barely changes their individual accuracy. On MMLU it removes information that receiver-local aggregation needs; on BoolQ, where the saboteurs carry almost no signal, it helps.

3 Setting and threat model

Swarm. N agents answer a stream of questions $t = 1, 2, \dots$. On each question every agent broadcasts one answer. Receiver i hears its own answer and a subset of its peers’ answers (visibility p_{obs}), and must output a final answer using only what it has heard on this and earlier questions. No labels are ever revealed. A fraction f of the agents is Byzantine; the rest are honest. Every honest agent is a receiver, and we score the mean accuracy over honest receivers. Byzantine receivers are never scored.

What the receiver knows. One thing: it is itself honest, i.e. it reports its own best answer. It does not know f , who the attackers are, which model produced a peer’s answer, or any label. Self-reported confidence is ignored throughout. It can be forged, and the original study showed a falsified self-report falls below chance [8].

Adversary. Byzantine agents may (i) know the gold answer, (ii) be *rushing*, seeing the honest broadcasts on the current question before speaking, (iii) coordinate arbitrarily with each other, (iv) know the defence and its published constants, and (v) change behaviour over time. Table 1 lists the strategies we evaluate. They are grouped by the assumption of our method each one attacks. Nine are symbolic. Four replay answers that real LLMs wrote when prompted to deceive [8]. Three more (Section 8) come from live inference in this work.

4 Receiver-anchored channel estimation

4.1 Model

For receiver i and question t , let $\mathcal{S}_t$ be the candidate set: the label space for closed questions (C options), or the distinct answers heard on t for open questions. Two open answers are merged when an equivalence checker says they denote the same value. This needs no gold (Section 6). Let $K_t = |\mathcal{S}_t|$. RACE fits the symmetric one-coin latent-truth model

$$Y_t \sim \text{Unif}(\mathcal{S}_t), \quad P(R_{tj} = Y_t) = a_j, \quad P(R_{tj} = c \mid Y_t \neq c) = \frac{1 - a_j}{K_t - 1}, \quad (1)$$

with one accuracy a_j per channel, including the receiver’s own. Channels are conditionally independent given Y_t , up to the clone tempering described below. The posterior score of candidate c is

Table 1: Attack zoo. “Rushing” attacks see the honest answers before speaking. The last column names the assumption of receiver-anchored estimation that the attack targets. Theorem 1 shows the minimax stationary attack is *uninformative/echo*.

Attack	Behaviour	Targets
coherent	one shared wrong answer per question (the original threat)	plurality
independent	each attacker lies on its own; never coordinates	coherence gates
gate-aware(p)	shares the lie w.p. p , else lies alone; p tuned below a published ceiling	coherence gates
partial(e)	coherent lie w.p. e , else the truth	estimation
uninformative	uniform random answer, independent of the truth	(minimax)
echo	copies a random honest answer ($B \perp Y \mid H$)	(minimax)
attractor	always backs the honest agents’ most common wrong answer (rushing)	independence
camouflage(θ)	truthful unless the honest vote is contested (share $< \theta$), then backs the strongest wrong honest answer (rushing)	independence
sleepers	replays an honest model on history, lies coherently afterwards	stationarity
LLM (replayed)	answers written by Llama-3.2-3B and Ministral-3-14B under four deception prompts: always-wrong, rushing, semantic hallucination, semantic negation	realism
LLM (live)	six small open models as covert saboteurs: solo, rushing, colluding (Section 8)	realism

a *signed* weighted vote:

$$s_t(c) = \sum_j w_{tj} \mathbf{1}[R_{tj} = c] \lambda_{tj}, \quad \lambda_{tj} = \log \frac{(K_t - 1) a_j}{1 - a_j}. \quad (2)$$

A report with $a_j > 1/K_t$ is evidence *for* the reported answer (TRUST). A report with $a_j < 1/K_t$ is evidence *against* it, spread over the alternatives (INVERT). $a_j = 1/K_t$ contributes nothing (DISCARD). AIP’s gate is a three-level quantisation of Equation (2). Unlike AIP, RACE decides the sign from a_j , not from how coherent a channel is (Proposition 5).

4.2 Anchored EM

The parameters are fitted by EM [10] on the receiver’s own unlabelled history. Label-free latent-class models are identified only up to a relabelling of the latent truth (Proposition 3). When a coherent bloc holds a plurality, the likelihood prefers the relabelling in which the liars are the experts. RACE removes the ambiguity with the receiver’s self-knowledge, in three ways:

1. **Anchor initialisation.** The first E-step puts posterior mass 0.75 on the receiver’s own answer, not on the plurality.
2. **Anchor prior.** $a_i \sim \text{Beta}$ with mean 0.75 and strength 8. Peers get a weak prior centred at chance (strength 2), so an unfamiliar peer starts at DISCARD, not TRUST.
3. **Anchor constraint.** After every M-step a_i is projected to $\geq 1/\bar{K} + 0.05$, where $1/\bar{K}$ is the chance rate on the tasks the receiver answered.

The M-step is the MAP update $a_j = (\sum_t \omega_t q_t(R_{tj}) + \kappa_j m_j) / (\sum_t \omega_t + \kappa_j)$, where q_t is the current posterior and ω_t an optional task weight (exponential forgetting in the online variant). The E-step

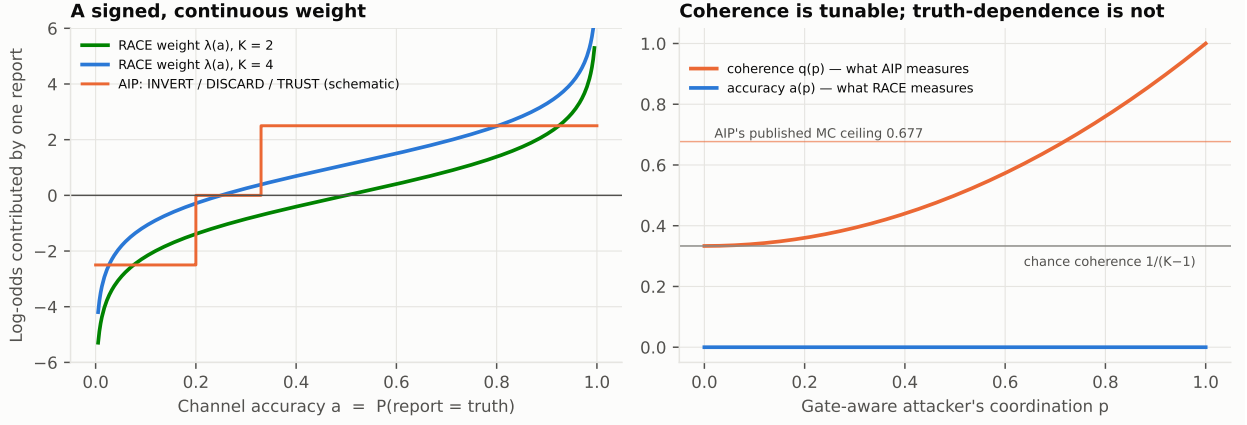


Figure 1: Left: RACE’s weight $\lambda(a)$ is signed and continuous; AIP’s gate (schematic) quantises it into three levels. Right: the gate-aware attacker never states the truth ($a = 0$, so $\lambda = -\infty$) at every coordination level p , while its pair coherence $q(p)$ sweeps from chance to 1 and crosses AIP’s published ceiling. Coherence is under the attacker’s control; dependence on the truth is not.

is Equation (2) followed by a softmax. Accuracies are clipped to $[10^{-3}, 1 - 10^{-3}]$. At prediction time unseen agents carry zero weight, and ties go to the receiver’s own answer.

4.3 Binary questions: class-conditional channels (v3.1)

On binary questions RACE replaces the one-coin channel with a class-conditional one, the two-coin model of Raykar et al. [35]: each channel j has a 2×2 confusion matrix $\pi_j[y, r] = P(R_{tj} = r \mid Y_t = y)$, and the truth has a class prior ρ . Each channel then has two parameters rather than one, and a short history identifies them. The anchor carries over. The receiver’s confusion matrix gets a Dirichlet prior with diagonal mean 0.75 and strength 8, peers get a flat Dirichlet prior of strength 2, and every diagonal entry of the receiver’s matrix is kept $\geq 1/2 + 0.05$ after each M-step. This is the row-diagonal dominance that makes the anchor identifying in Proposition 3(c). A report r adds $w_{tj} \log(\pi_j[r, r]/\pi_j[\bar{r}, r])$ to the log-odds of r , where $\bar{r}$ is the other answer. We summarise a channel by half its log diagnostic odds ratio,

$$\lambda_j = \frac{1}{2} \log \frac{\pi_j[0, 0] \pi_j[1, 1]}{\pi_j[0, 1] \pi_j[1, 0]}, \quad (3)$$

which is positive for an informative channel, negative for an inverter and close to 0 for a channel that says “yes” regardless of the truth. The one-coin model cannot tell that last channel from a coin flip, and it mis-weights channels that are informative only when they answer against their bias. For more than two options the one-coin model is kept. A $K \times K$ matrix per channel is over-parameterised for the histories we use. Averaged per benchmark, the full-confusion variant is never more than 0.1 points better than RACE on four-option questions in any study, and it is far worse in the live swarm.

Provenance. This rule is an amendment. RACE v3.0 used the one-coin model for every answer space and was frozen before the confirmation sets were evaluated. The live swarm (Section 8) showed that small models answer binary questions with a strong option bias, and that v3.0 fell below the receiver alone there. v3.1 differs from v3.0 only on binary questions. Every study was re-run with v3.1, and v3.0 is reported alongside it as `race_onecoin`. On the replayed BoolQ worlds

Algorithm 1 RACE fit for one receiver i (history $\mathcal{H}$, no labels)

- 1: Encode reports R_{tj} over candidate sets $\mathcal{S}_t$; $K_t \leftarrow |\mathcal{S}_t|$
 - 2: $G \leftarrow$ complete-link groups of same-source links (raw agreement ≥ 0.99)
 - 3: $(a, q) \leftarrow \text{ANCHOREDEM}(R, G)$ $\triangleright$ init at i 's answers; Beta prior; $a_i \geq 1/\bar{K} + \delta$
 - 4: $\hat{y}_t \leftarrow \arg \max_c q_t(c)$
 - 5: $G \leftarrow$ complete-link groups of same-source $\cup$ dependent-error links w.r.t. $\hat{y}$
 - 6: $(a, q) \leftarrow \text{ANCHOREDEM}(R, G)$
 - 7: **return** a, G $\triangleright$ predict with Equation (2), tempering within G
-

of E1–E3 the two versions differ significantly in 1 of 78 cells, always in favour of v3.1 (Table 4). The live BoolQ result for v3.1 is therefore not a confirmation.

4.4 Clone tempering

A swarm usually replicates models. A coordinated bloc is, by construction, many copies of one lie. Under conditional independence both are over-counted. RACE groups channels and tempers each report by $w_{tj} = 1/(\text{number of heard members of } j\text{'s group on } t)$, so a group contributes one unit of evidence per question. Groups are complete-link clusters over two kinds of link:

- **Same-source links:** raw agreement ≥ 0.99 over co-observed history, i.e. essentially identical answer streams. These are applied *before* any fit, so a receiver's own replica can never be counted as independent confirmation of the receiver.
- **Dependent-error links:** agreement ≥ 0.95 restricted to history tasks where at least one member is wrong relative to a first-stage truth estimate, with at least 8 such tasks.

Raw agreement is the wrong statistic for the second kind: two distinct but accurate models agree on almost every task simply because both are usually right (Proposition 7).

Cost. One EM iteration is $O(|\mathcal{H}| \cdot N \cdot K)$. With 80 history tasks and 10 agents a fit converges in 10–20 iterations and takes a few milliseconds. The live-swarm run (Section 8) spends about three orders of magnitude more time on generating answers than on aggregating them.

Hyper-parameters. All were fixed before any evaluation data were inspected and were never tuned: anchor mean 0.75, anchor strength 8, anchor margin 0.05, peer strength 2, thresholds 0.99/0.95, minimum support 8, clip 10^{-3} . Two design decisions were revised after inspecting development data. Both revisions are recorded in Appendix A, and the superseded variants appear in every table. The method was then frozen as v3.0, and two confirmation sets that had not been inspected were evaluated (Sections 7.3 and 8). The binary rule of Section 4.3 (v3.1) was added after the second confirmation set, and it is reported as such.

5 Theory

Let Y be the truth on a question, H the honest reports (including the receiver's), and B the Byzantine reports. An adversary strategy σ is any conditional law $B \sim \sigma(\cdot \mid Y, H)$, so the adversary may be gold-aware and rushing.

Theorem 1 (A lie can only add information; the minimax attack is silence). *Let $\text{acc}^*(\sigma) = \mathbb{E}[\max_y P_\sigma(Y=y \mid H, B)]$ be the accuracy of the Bayes receiver that knows the joint law induced by σ , and $\text{acc}_H^* = \mathbb{E}[\max_y P(Y=y \mid H)]$. Then $\text{acc}^*(\sigma) \geq \text{acc}_H^*$ for every σ , with equality whenever*

$B \perp Y \mid H$. Hence $\inf_{\sigma} \text{acc}^*(\sigma) = \text{acc}_H^*$, and the infimum is attained, for example by answering at random or by echoing a random honest report.

Proof. $P(Y=y \mid H) = \mathbb{E}[P(Y=y \mid H, B) \mid H]$. The maximum is convex, so by Jensen $\max_y \mathbb{E}[\cdot \mid H] \leq \mathbb{E}[\max_y \cdot \mid H]$. Take expectations. If $B \perp Y \mid H$, the two posteriors coincide. $\square$

Corollary 2 (No breakdown point for a known-channel receiver). *Since acc_H^* is at least the receiver’s own accuracy, the known-channel receiver never falls below “trust yourself”, at any $f < 1$. By contrast, any plurality rule reaches 0 against a coherent bloc once $f > 1/2$.*

Theorem 1 is Blackwell’s comparison of experiments [3] read adversarially. Its consequence is that robustness is an *estimation* problem. The adversary has exactly three levers:

1. estimation error;
2. violating the model’s assumptions, namely conditional independence given Y and stationarity;
3. withholding its own information.

Section 7 is organised around these levers.

Proposition 3 (Label switching, and what the anchor buys). *(a) Binary one-coin model. For Y uniform on $\{0, 1\}$ and conditionally independent reports with accuracies $\theta = (a_1, \dots, a_n)$, the report law under θ equals the law under $\theta' = (1 - a_1, \dots, 1 - a_n)$. If a coherent always-wrong bloc holds a majority, any rule that selects the labelling in which the plurality is accurate outputs the lie on every question. This covers majority vote, plurality-initialised EM and label-free Dawid–Skene. (b) Dawid–Skene. For any permutation τ of latent classes, relabelling the class prior and the rows of every confusion matrix by τ leaves the report law unchanged. Uniqueness results [1, 24] identify the model only up to such τ . (c) Anchor. If the receiver’s confusion matrix is strictly row-diagonally dominant, $\pi_i[k, k] > \pi_i[k, l]$ for all $l \neq k$, then among all relabellings of the true parameters only $\tau = \text{id}$ keeps π_i row-diagonally dominant.*

Proof of (c). For $\tau \neq \text{id}$ pick k with $m = \tau^{-1}(k) \neq k$. Row k of the relabelled matrix is row m of π_i . Its largest entry is in column $m \neq k$, so dominance fails in row k . $\square$

The anchor is also *necessary*. In the binary case with $a_i = 1/2$, θ and θ' both satisfy any constraint the truth satisfies. A near-chance receiver therefore cannot identify the truth from its own history, whatever estimator it uses. We observe exactly this weak-anchor capture on open-answer questions (Section 9).

Theorem 4 (Anchored estimation is identified and consistent). *Consider the class-conditional model: questions i.i.d. with a class prior ρ of full support on K classes, and $G \geq 3$ channel groups that are conditionally independent given Y , each group counted once after clone tempering. Suppose three of the groups have invertible confusion matrices; for a one-coin channel this means accuracy $a \neq 1/K$, so a consistent liar counts. Suppose also that the receiver’s confusion matrix is row-diagonally dominant with margin $\delta > 0$, i.e. $\pi_i[k, k] \geq \pi_i[k, l] + \delta$ for all $l \neq k$. Then:*

- (a) *the parameters $(\rho, \pi_1, \dots, \pi_G)$ are identified from the law of the reports, exactly and not merely up to a relabelling of the classes, within the set Θ_δ where the receiver is δ -dominant;*
- (b) *the maximum-likelihood estimator over Θ_δ , with entries clipped to $[\epsilon, 1 - \epsilon]$, is strongly consistent as the history length $T \rightarrow \infty$, and so is the anchored MAP estimator, whose fixed-strength priors vanish relative to T ;*
- (c) *the plug-in decision converges almost surely to the decision of the known-channel receiver, so the accuracy of the anchored estimator converges to acc^* of Theorem 1 under the model.*

For binary questions this means three channels that are informative in either direction, i.e. whose “yes” rates differ between yes- and no-questions.

Proof. (a) The joint law of the three groups is a three-way array $\sum_y \rho_y \pi_1[y, \cdot] \otimes \pi_2[y, \cdot] \otimes \pi_3[y, \cdot]$ whose factor matrices have full Kruskal rank K , and $3K \geq 2K + 2$. By Kruskal’s uniqueness theorem [24], as applied to latent-class models by Allman et al. [1], it determines ρ and the three confusion matrices up to one simultaneous permutation τ of the classes; the row-stochastic normalisation fixes the scales. Every further group g is then identified from its joint law with group 1, $\pi_1^\top \text{diag}(\rho) \pi_g$, because π_1 and $\text{diag}(\rho)$ are invertible. By Proposition 3(c) only $\tau = \text{id}$ keeps the receiver dominant. (b) Θ_δ with clipping is compact, the log-likelihood is continuous in the parameters, and by (a) the true parameter is its unique maximiser in expectation, so the standard consistency argument for maximum likelihood applies. The Beta and Dirichlet priors have fixed strength and are therefore $O(1/T)$ relative to the likelihood. (c) The posterior is continuous in the parameters, and the Bayes decision changes only on a set of reports of measure zero. $\square$

The theorem is about the estimator, not the algorithm. EM converges to a stationary point, and the anchor initialisation is what places it in the basin of the dominant solution. It has two further consequences. First, identification needs three informative groups. With only two, as in a swarm of the receiver plus one peer, the binary class-conditional model is not identified even with an anchor, and our unit tests reproduce this. Second, if the model is misspecified, the estimator converges to the Kullback–Leibler projection of the truth onto the model. This is what happened to the one-coin model on option-biased live BoolQ answers (Section 8), and it is why v3.1 uses class-conditional channels on binary questions (Section 4.3).

Proposition 5 (AIP’s gate quantises the Bayes weight; Proposition 1 of [8] does not bind RACE). *Under the one-coin model a report $R_j = c$ shifts the log-odds of c against any alternative by $\lambda_j = \log \frac{(K-1)a_j}{1-a_j}$, and $\text{sign } \lambda_j = \text{sign}(a_j - 1/K)$. Consider the gate-aware attacker: it joins a shared lie with probability p , otherwise lies independently, and never states the truth. Its pair coherence is $q(p) = p^2 + (1 - p^2)/(K - 1)$, which sweeps $[1/(K - 1), 1]$. Yet $a_j = 0$ and $\lambda_j = -\infty$ for every p .*

Coherence therefore carries no information about the quantity that determines the Bayes weight (Figure 1). The original impossibility result concerns rules measurable with respect to per-question coherence. RACE’s statistic depends on the joint law of (R_j, Y) , estimated through the anchored latent model, so it lies outside that class.

Proposition 6 (History needed to invert correctly). *Fix the groups. Let $\hat{y}$ have error rate δ on the n_j history questions where peer j was heard, and let $\hat{a}_j$ be j ’s agreement with $\hat{y}$. Then*

$$P(\text{sign}(\hat{a}_j - 1/K) \neq \text{sign}(a_j - 1/K)) \leq 2 \exp(-2n_j(|a_j - 1/K| - \delta)^2) \quad \text{whenever } |a_j - 1/K| > \delta.$$

Proof. $|\hat{a}_j - \bar{a}_j| \leq \delta$, where $\bar{a}_j$ is the empirical accuracy. Apply Hoeffding’s inequality [15] to $\bar{a}_j$. $\square$

Take an always-wrong liar on four-way questions ($|a_j - 1/4| = 0.25$) and $\delta = 0.1$. Then $n_j = 80$ gives a bound of 0.055, and $n_j = 160$ gives 0.003. Figure 9 measures the realised curve.

Proposition 7 (Raw agreement confuses competence with dependence). *Take two channels that are conditionally independent given Y , with accuracies a, b and symmetric errors. Then $P(R_1 = R_2) = ab + \frac{(1-a)(1-b)}{K-1} \rightarrow 1$ as $a, b \rightarrow 1$, while*

$$P(R_1 = R_2 \mid R_1 \neq Y \text{ or } R_2 \neq Y) = \frac{(1-a)(1-b)}{(K-1)(1-ab)} \leq \frac{1}{K-1}.$$

For replicas of one source, and for a perfectly coordinated bloc, the second quantity equals 1.

Proof. $P(\text{equal, not both correct}) = (1 - a)(1 - b)/(K - 1)$ and $P(\text{not both correct}) = 1 - ab \geq (1 - a)(1 - b)$. $\square$

The same confound explains a refuted recipe in the original ledger: “maximise effective swarm size” largely restated roster weakness [8].

6 Experimental protocol

Data. We use the X3-repaired inference cache of the original study [8]: greedy answers of nine open-weight models on six benchmarks. The benchmarks are MMLU [13], MedQA [20] and ARC [6] (four options each), BoolQ [5] (binary), and GSM8K [7] and MATH-500 [14, 28] (open answers). That is 200 questions per benchmark and 500 for GSM8K. The frozen roster has seven models from seven labs, ranging from 44.5% (Llama-3.2-3B) to 94.0% (Gemma-4-31B) on MATH-500. The cache also contains answers that Llama-3.2-3B and Ministral-3-14B wrote under four deception prompts. Before the repair, 944 of 13,500 rows were answers “manufactured” by the extractor from truncated generations (35% of MATH-500 rows). All earlier MATH-500 conclusions are therefore superseded (Section 10).

Worlds. A world has a benchmark, a Byzantine fraction f , an attack, a visibility p_{obs} and a seed. It contains $N = 10$ identities. Each honest identity replays one cached model, round-robin over a shuffled roster, so some models repeat. Byzantine identities are a prefix of one fixed permutation, which nests the corrupted sets across f . Questions are split by a stable hash into disjoint HISTORY / VALIDATION / TEST partitions (40/20/40%). Every receiver fits on HISTORY only and is scored on TEST. VALIDATION is used only to select attacks for the best-response analysis. The defence input is stripped of gold, Byzantine flags, model names and attack labels.

Open answers. Two answers on the same question are merged when the repository’s equivalence checker (symbolic `math_match` for MATH-500, numeric equality for GSM8K) says they denote the same value. This compares answers with *each other*, never with the key, so a deployed receiver can do the same. It is applied identically to every method.

Methods. *Deployable*: receiver alone; majority vote; logprob confidence weighting; SAC filter-refine; AIP gated, trust-only, naive and soft-gate (M1); one-coin label-free Dawid–Skene (plurality-initialised, the unanchored ablation of RACE); full Dawid–Skene [9]; and RACE with its variants (full confusion, self-cap, and three superseded variants kept as ablations). *Oracles* (upper references, not deployable): majority over the honest agents a receiver heard, and a known-channel decoder that uses RACE’s scoring rule with accuracies measured against gold on HISTORY. All AIP variants use the published calibrated ceilings and the fixed code of [30].

Studies.

- **E1 (main):** $f \in \{0, .1, \dots, .9\}$, coherent liars, $p_{\text{obs}} \in \{1, .5\}$, 5 seeds; 600 worlds.
- **E2 (attack zoo):** 14 symbolic attack settings at $f \in \{.3, .5, .7\}$, 3 seeds; 756 worlds.
- **E3 (replayed LLM deceivers, confirmation set):** 4 prompts, $f \in \{.1, .3, .5, .7\}$, 5 seeds; 480 worlds.
- **E4 (composition, replication, swarm size).**
- **E5 (history length).**
- **E6 (online, predict-then-commit, under changing attackers).**

Coherent liars: plurality and label-free rules break at $f = 1/2$, AIP at 0.9 (BoolQ: 0.5); RACE holds

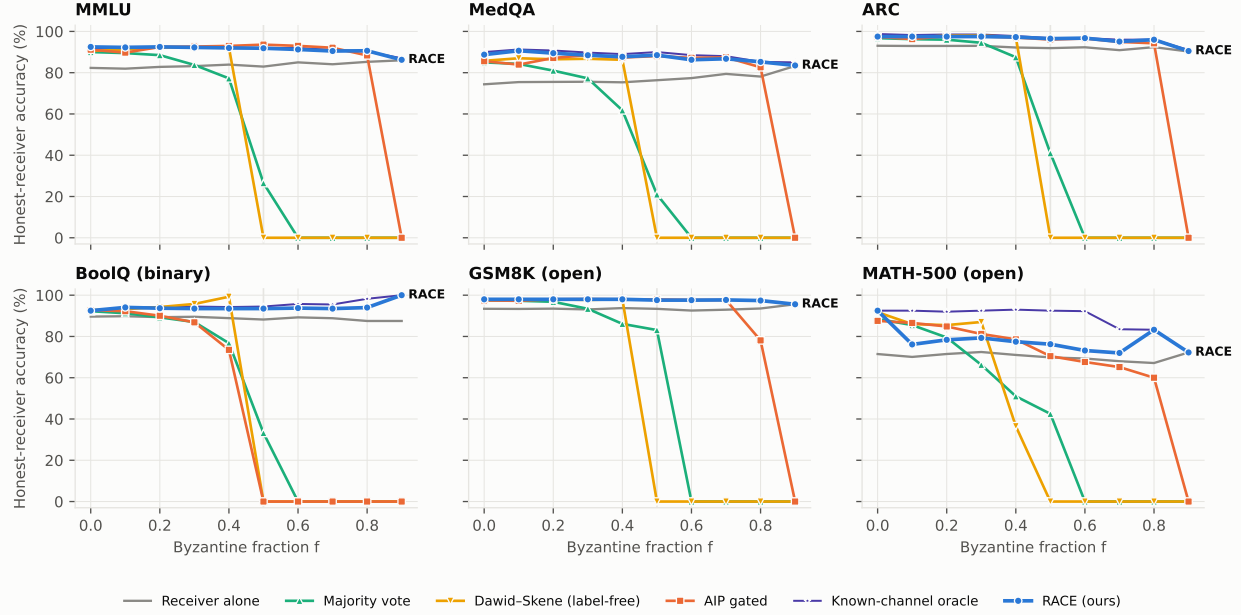


Figure 2: E1. Honest-receiver accuracy against a coherent lying bloc, as a function of the Byzantine fraction f (TEST split, 5 seeds, full visibility). Majority and label-free Dawid-Skene fall to 0 at $f \geq 1/2$ (label switching, Proposition 3). AIP collapses on binary BoolQ at $f = 0.5$ and everywhere at $f = 0.9$. RACE never falls below the receiver alone at any f , except on MATH-500 at small f (weak-anchor capture, Section 9).

- **E7 (channel-estimation diagnostics).**
- **E8 (live six-model swarm, confirmation set).**

Statistics. Receivers, seeds and peers within a world reuse the same cached answers, so they are not independent samples. We average accuracy over honest receivers and seeds within each question. We then compare methods on the same questions, using a paired percentile bootstrap (2,000 resamples) for 95% intervals and a Wilcoxon signed-rank test for p -values. Within each study, p -values are Holm-corrected [16] over all cells and baselines. A verdict is “win”/“loss” only if the corrected $p < 0.05$ and the bootstrap interval of the mean difference excludes zero; the two criteria disagree in none of the per-benchmark cells of E1–E3, but they can when cells pool heterogeneous benchmarks.

Compute. Everything except E8 replays cached answers on a 4-core CPU without a GPU. E8 runs fresh inference with six 1–2B-parameter models on the same CPU. Wall time and peak memory are recorded in each study’s manifest.

7 Results

7.1 E1: no breakdown point against a coherent bloc

Figure 2 is the headline. Averaged over the six benchmarks at $f = 0.5$, RACE reaches 90.7% against 74.3% for AIP, 41.2% for majority vote, 0.0% for label-free Dawid-Skene and 83.8% for the receiver

Attack zoo (6 benchmarks): blue = pooling helps, red = pooling hurts

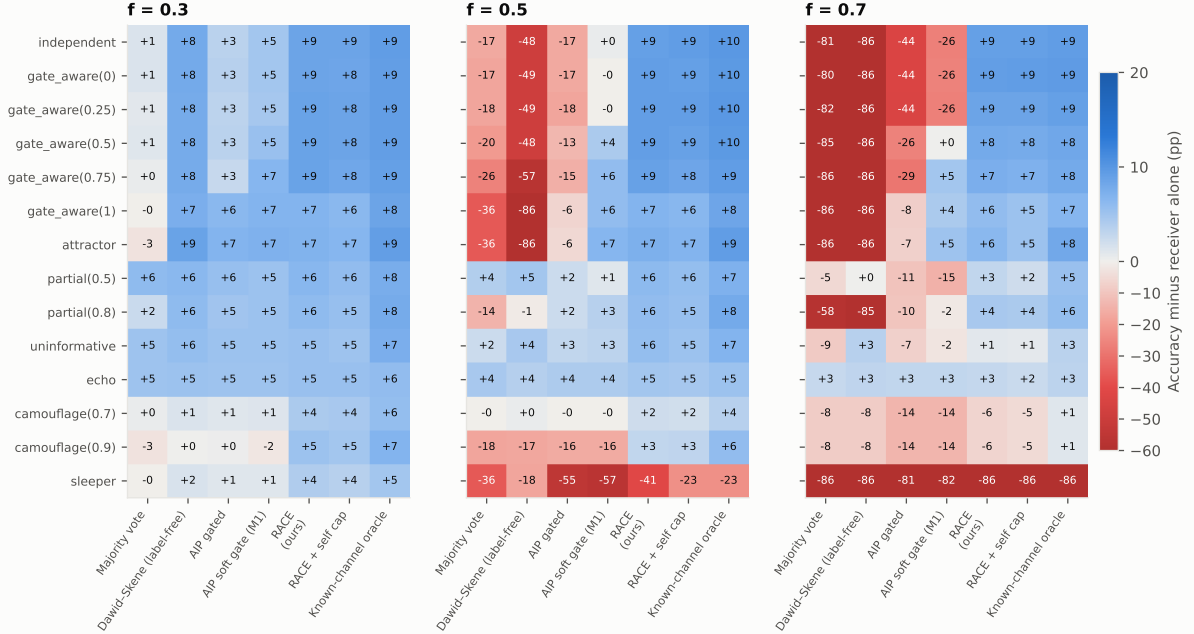


Figure 3: E2. Accuracy minus the receiver alone (percentage points), averaged over six benchmarks and three seeds, for 14 attack settings at three Byzantine fractions. Blue: pooling helps. Red: pooling hurts.

alone. At $f = 0.9$ the figures are 88.0%, 0.0%, 0.0%, 0.0% and 85.8%. The mean accuracy of majority vote and label-free Dawid-Skene first falls below half the receiver’s at $f = 0.5$ and $f = 0.5$. For AIP this happens at $f = 0.9$, and for RACE at none of the tested fractions. The known-channel oracle bounds the remaining headroom at 93.9% ($f = 0.5$).

The binary case, which AIP cannot handle, is where inversion is most valuable. On BoolQ at $f = 0.7$, RACE scores 93.5% (AIP: 0.0%, receiver: 88.8%). A bloc that always says the opposite is a perfect oracle once it is read backwards. With all agents honest ($f = 0$), RACE costs nothing: 93.6% against 92.0% for AIP and 91.7% for majority. Across all 6×10 cells (Holm-corrected Wilcoxon, Table 2), RACE is significantly better than AIP in 20 cells and worse in 2. Against majority it wins 41 cells and loses 1. Against the receiver alone it wins 37 and never loses. Every significant loss is on MATH-500 at $f \leq 0.3$, the weak-anchor regime analysed in Section 9. Clone tempering has a price against a *perfectly* coherent bloc. Counting the bloc’s identical vetoes separately happens to help there (93.8% without tempering at $f = 0.5$), but the tempered rule is the one that wins on the confirmation sets (Section 7.3).

7.2 E2: the attack zoo and the gate-aware adversary

Figure 3 organises the attacks by the assumption they target. Three findings stand out.

Coherence gates are blind to incoherent liars. Against the gate-aware attacker at $f = 0.7$ (coordination $p \leq 0.5$, i.e. below AIP’s published ceiling), AIP scores 48.5% and its soft gate 68.9%, while RACE scores 94.9% (Figure 4). Independent liars, who never coordinate at all, give 42.8% versus 95.3%. This is Proposition 5 in data. The attacker’s coherence is tunable, but its dependence on the truth is not.

The minimax attack costs little. Uninformative liars are the minimax strategy of Theorem 1. Against them RACE stays at 87.3% at $f = 0.7$, close to the honest-only oracle (89.5%) and above

The gate-aware adversary: AIP is defeated below its ceiling; RACE's weight never depended on coherence

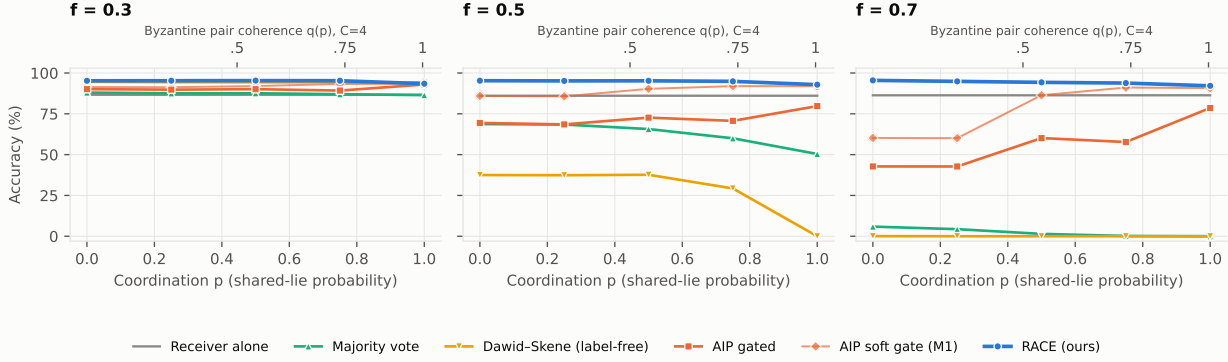


Figure 4: The gate-aware adversary. The upper axis shows the attacker’s pair coherence $q(p)$. AIP is defeated exactly in the sub-ceiling band. RACE is nearly flat in p (within about 3.5 points at $f = 0.7$), because the attacker’s accuracy, not its coherence, sets the weight.

Adaptive attacker: best response per defender, benchmark, f and seed

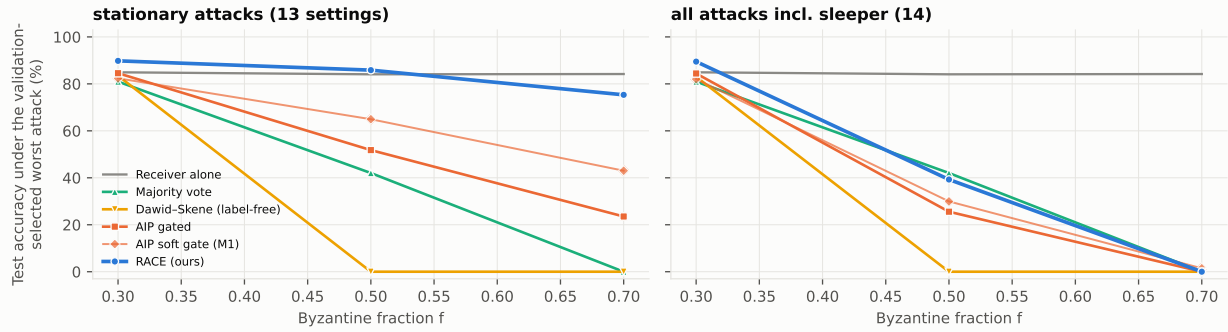


Figure 5: Best-response attacker. The attack is selected on VALIDATION for each defender separately and scored on TEST.

the receiver alone (86.4%).

RACE’s own failure modes. Camouflage (Section 9) is the one stationary attack that pushes RACE below the receiver at $f = 0.7$ (80.1% vs 86.4%). The sleeper defeats every static fit.

Adaptive attackers. For every defender, benchmark, f and seed we let the attacker pick, on VALIDATION, whichever stationary zoo attack hurts that defender most, and score TEST under that choice (Figure 5). Under its own best response, RACE holds 85.8% at $f = 0.5$ and 75.3% at $f = 0.7$. The corresponding figures are 51.8% and 23.5% for AIP, 42.0% and 0.0% for majority, and 84.1% and 84.2% for the receiver alone. Including the sleeper, every static method fails at $f \geq 0.5$. Section 7.9 addresses this with online refitting.

7.3 E3: replayed LLM deceivers (confirmation set)

The four prompt-injected attacks of the original study were written by real LLMs. Their lies are neither perfectly wrong nor perfectly coordinated. This set was not inspected before the method was frozen. RACE is above every non-RACE method at every f , and within 0.1 points of its own best

Replayed real-LLM deceivers (4 prompts $\times$ 2 attacker models $\times$ 6 benchmarks)

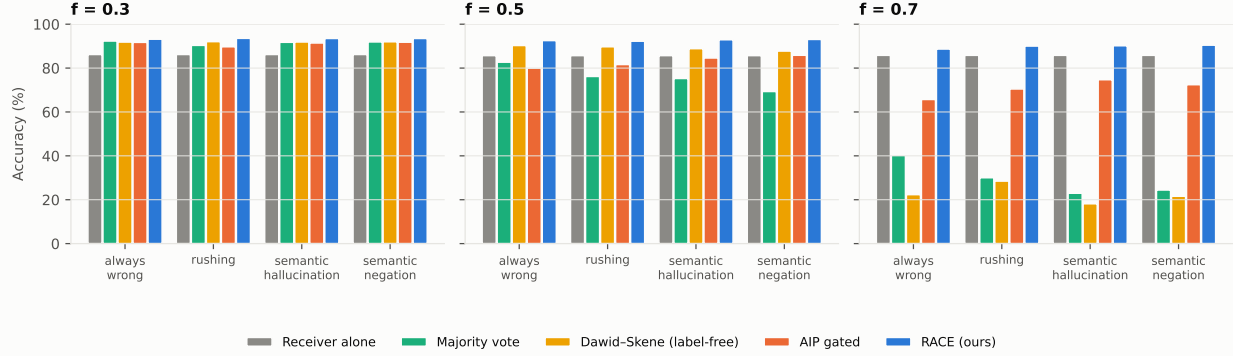


Figure 6: E3. Replayed real-LLM deceivers (Llama-3.2-3B and Ministral-3-14B under four deception prompts). Confirmation set, evaluated after the method was frozen.

variant (Figure 6). At $f = 0.7$ it scores 89.8%, against 70.8% for AIP, 29.4% for majority, 22.7% for Dawid-Skene and 85.8% for the receiver alone. That is within 91.1% of the known-channel oracle and above the honest-only oracle (88.7%), because inverting the liars adds information that honest agents alone do not have. Across 480 Holm-corrected comparisons (5 baselines $\times$ 6 benchmarks $\times$ 4 prompts $\times$ 4 fractions), RACE has 197 significant wins and 0 losses. The confirmation result proper is that of the frozen v3.0 rule, which differs from v3.1 only on BoolQ: 196 wins and 0 losses. Both superseded variants of the clone rule score lower on this set (89.1% and 84.0% at $f = 0.7$). This is independent support for the one post-hoc design change (Appendix A).

7.4 E7: does RACE estimate the channels?

RACE’s estimates are label-free. We compared them with each peer’s true accuracy on HISTORY, which only the evaluator can compute, over 27,216 receiver-peer channels spanning seven attacks, four fractions and six benchmarks (Figure 7). The correlation is $r = 0.977$ and the mean absolute error is 0.032. RACE inverts 75.7% of Byzantine channels, against 49.2% for AIP. It falsely inverts 0.8% of honest channels (AIP: 1.5%). The Byzantine channels RACE trusts (19.9%) are liars that tell the truth often enough to be positively informative, such as partial and camouflage attackers and LLM liars that fail to lie.

Anchor strength. Figure 8 groups receivers by their own accuracy. For receivers above 50%, RACE ends more than 5 points worse than trusting themselves in only 2.1% of cases. The rates are 20.4% for AIP and 38.1% for majority. For receivers at or below 50%, the weak-anchor regime of Proposition 3, the rate rises to 24.1%.

7.5 E5: how much history?

RACE needs little history (Figure 9). Averaged over attacks and benchmarks at $f = 0.5$, it reaches 84.7% with 5 history questions and 91.0% with 20. AIP reaches 27.9% and 53.9%: its significance test has a minimum-support floor. This matches Proposition 6. An always-wrong liar is far from chance, so few joint observations separate it.

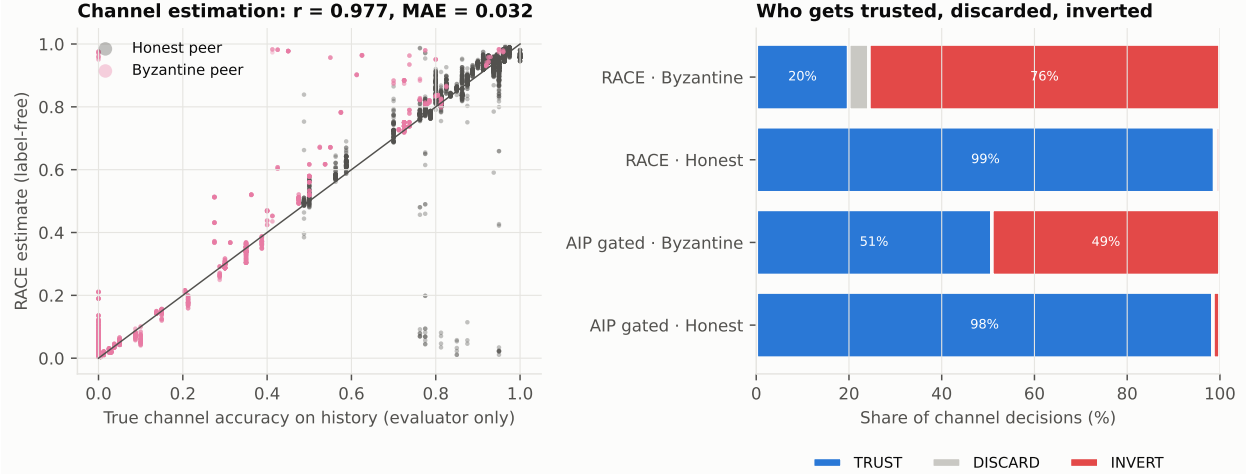


Figure 7: E7. Left: label-free RACE estimates versus true channel accuracy (evaluator-side). Right: share of TRUST/DISCARD/INVERT decisions for honest and Byzantine peers.

Anchor strength: RACE is almost never worse than self-trust once the receiver is above chance

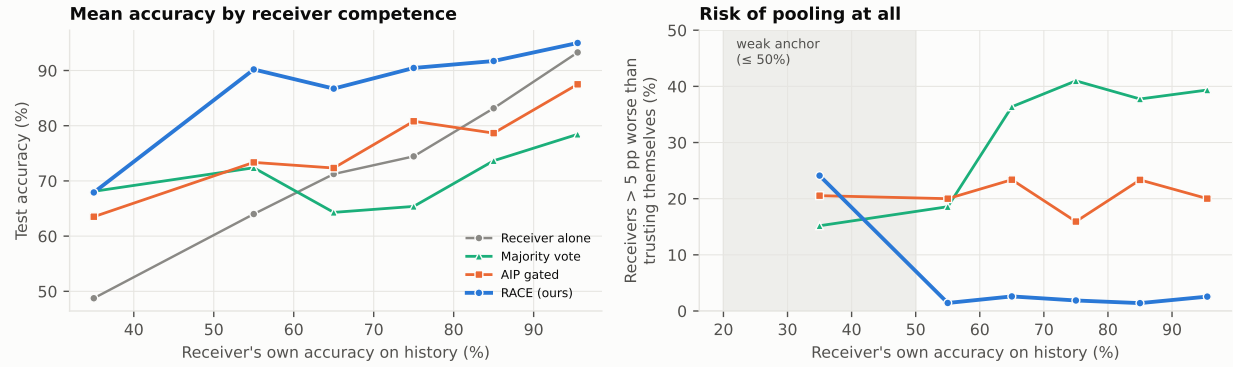


Figure 8: Receiver competence. Left: mean accuracy. Right: share of receivers that end > 5 points worse than trusting themselves.

7.6 E4: replication, composition and swarm size

Deployed swarms usually replicate one model (Figure 10). Take ten replicas of Llama-3.2-3B, the weakest roster model, facing a coherent bloc at $f = 0.5$. RACE scores 65.9% (AIP: 34.5%, receiver alone: 61.9%). With ten replicas of Qwen-3.8-27B the figures are 93.1%, 55.2% and 90.9%. In a homogeneous swarm the honest agents are one source, and clone tempering counts them once. RACE's gain over the receiver therefore comes entirely from what the liars leak: each lie removes one wrong option. Without tempering, the ten identical honest votes swamp that signal (61.9% for Llama, equal to the receiver alone). Adding the two weak-tier models to the roster ("7 + weak arm") gives 89.1% for RACE against 64.1% for AIP.

Swarm size helps RACE because larger swarms contain more *distinct* honest models. From 5 to 40 agents it rises from 84.6% to 89.5% against a coherent bloc, and from 88.3% to 96.4% against gate-aware liars (pooled over $f \in \{.3, .5, .7\}$). AIP moves from 37.8% to 78.7% and from 48.9% to 63.7%.

How much history does inversion need? (MMLU, MedQA, BoolQ, MATH-500)

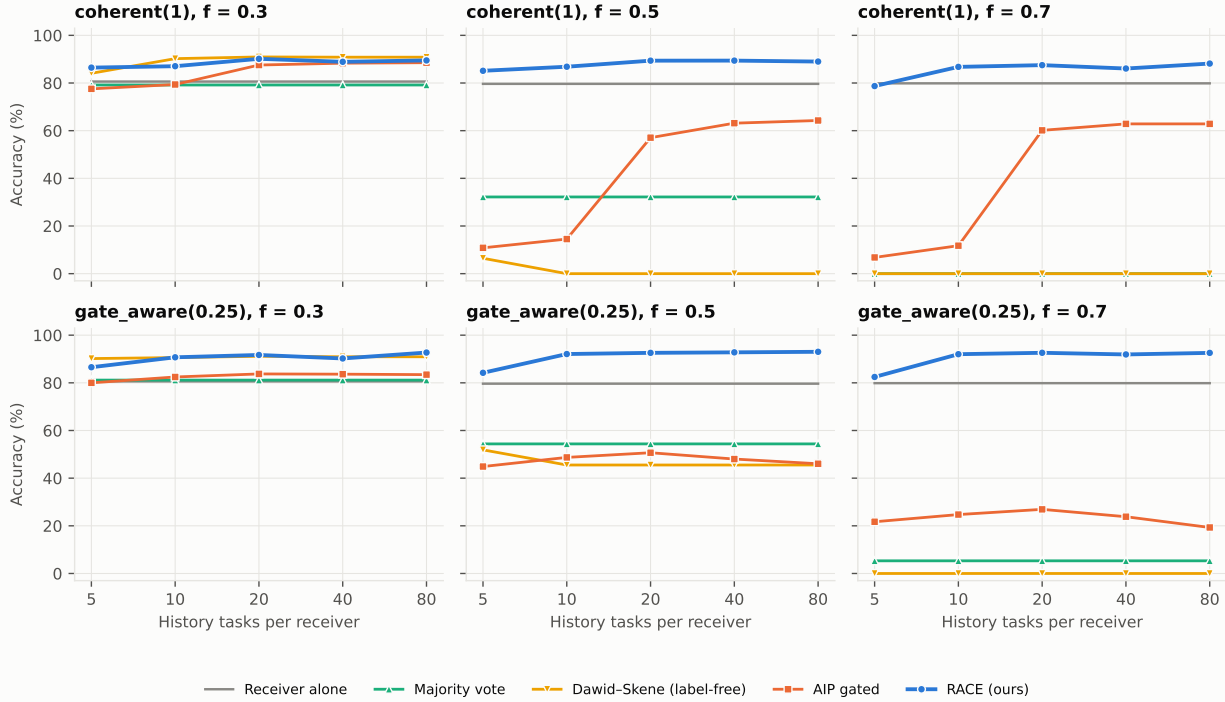


Figure 9: E5. Accuracy versus history length.

One trade-off recurs. On the weak four-model roster, clone tempering costs RACE accuracy against a perfectly coherent bloc: 65.1% versus 71.2% without tempering, with the known-channel oracle at 77.1%. Counting the bloc’s identical vetoes separately happens to help when the bloc is perfectly coordinated. The tempered rule is the one that generalises to real LLM deceivers (Section 7.3).

7.7 E9: what are the liars worth?

Theorem 1 says liars cannot remove information from a receiver that knows the joint law. It does not say how much information they add, or how much of it a label-free receiver can recover. We measure both with two oracles that see the gold labels of the HISTORY split. The *liar-removal oracle* knows who lies, removes every liar, and decodes the honest agents with their true accuracies and true source groups. The *known-channel oracle* keeps every agent, with true accuracies and true source groups. We call the gap between them the *information budget* of an attack. E9 evaluates both oracles, RACE and the extension below on 972 worlds: 14 symbolic and 4 LLM attack settings, $f \in \{0.3, 0.5, 0.7\}$, six benchmarks and three seeds. Figure 11 shows every method relative to the liar-removal oracle.

Truth-dependent liars are worth more than their absence. For independent and gate-aware liars the budget is positive and grows with f . For independent liars it is +1.8, +2.6 and +5.6 points at $f = 0.3$, 0.5 and 0.7. At $f = 0.7$, three honest agents plus seven liars decode better than the three honest agents alone. RACE captures most of this budget without labels and without knowing who lies. At $f = 0.7$ it scores 94.8% against independent liars, while the oracle

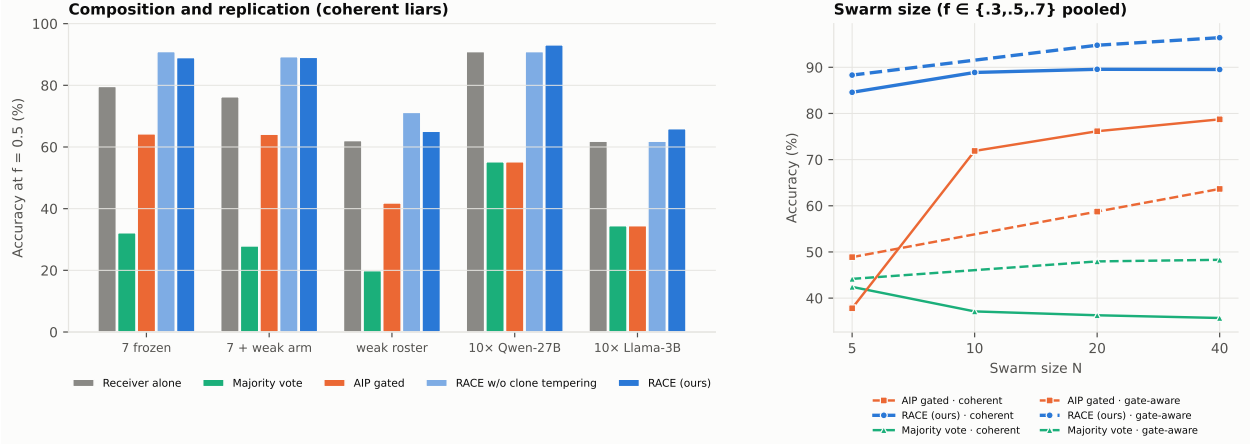


Figure 10: E4. Left: composition and replication at $f = 0.5$ (coherent bloc). Right: swarm size, by attack.

that removes every liar scores 89.7%. Pooling tasks over benchmarks, RACE is significantly *above* the liar-removal oracle in 16 of 54 (attack, f) cells. All of these cells are truth-dependent attacks: independent, gate-aware with $p < 1$ at every f , and the coherent bloc and attractor at $f = 0.7$. A rule that only filters suspected liars sees at most the honest reports, so in expectation it cannot beat the Bayes receiver on those reports, for which this oracle is a proxy. Exceeding it requires reading the liars.

LLM deceivers: the value is neutralisation, not extra information. For the four replayed LLM attacks the budget is close to zero: -0.1, -0.5 and -0.3 points on average. Deception prompts produce lies that are only weakly tied to the truth. At the one-coin level they carry little beyond what the honest agents already say. RACE’s contribution is then to cancel the damage. At $f = 0.7$, majority vote falls to 32.8% while RACE scores 88.0%. That recovers 70% of the gap between the receiver alone (84.2%) and the liar-removal oracle (89.7%); at $f = 0.5$ it recovers 86%. In the per-benchmark cells, RACE is statistically indistinguishable from the liar-removal oracle in 68 of 72 LLM cells and below it in 4.

A negative budget certifies a model violation. The known-channel oracle assumes conditionally independent, symmetric channels. An attack against which it does *worse* than the liar-removal oracle therefore lies outside that model. Three attacks do: echo (-1.8 at $f = 0.7$), camouflage (-4.1) and the sleeper (-34.1 at $f = 0.5$). The echo attacker copies honest reports, so its errors are not independent given the truth. The camouflage attacker conditions on the honest vote. The sleeper is non-stationary. These are exactly RACE’s failure modes (Section 9). An evaluator with labels can therefore use the sign of the budget to check whether a new attack is inside the model class that RACE assumes. Uninformative liars have a budget of essentially zero (-1.6 at $f = 0.7$), as Theorem 1 predicts for the minimax attack. RACE pays a small estimation cost there (-4.7 points relative to the liar-removal oracle), which is the price of not knowing who is silent.

A negative result: conditioning on disagreement (Race-D). Camouflage liars earn trust on questions where every honest agent agrees. A natural fix is to fit the channels only on history questions where the heard answers disagree, falling back to all questions when fewer than five qualify. We call this RACE-D. On the 14 attack settings other than camouflage, echo and sleeper,

What the liars are worth: accuracy relative to an oracle that removes every liar

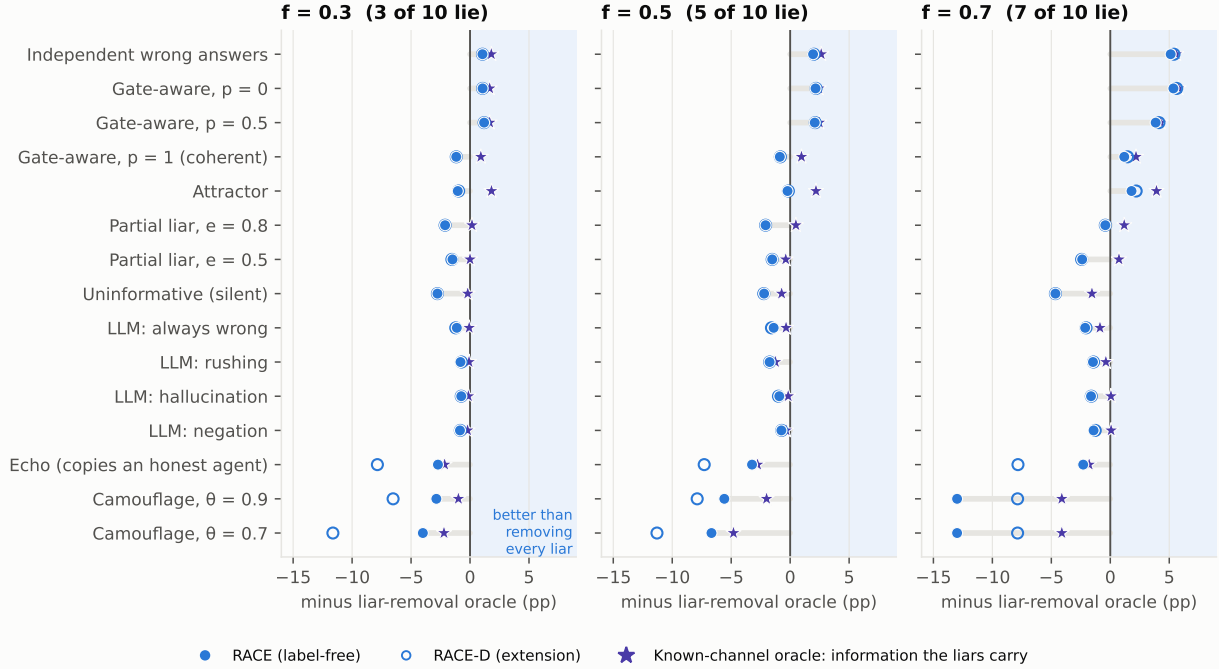


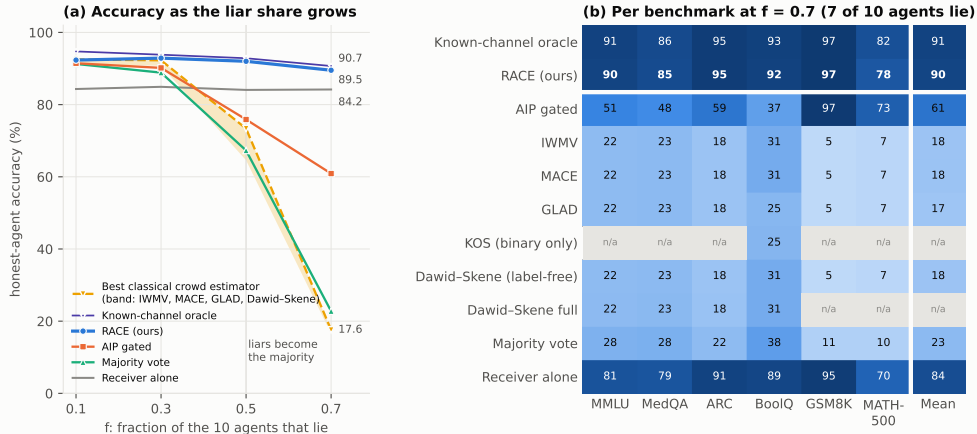
Figure 11: E9. What the liars are worth. Accuracy minus the liar-removal oracle, which knows who lies, removes them, and decodes the honest agents with their true accuracies. Violet stars: the known-channel oracle, which keeps the liars; its offset is the information budget. Filled blue: RACE. Hollow blue: RACE-D. Shaded: better than removing every liar. Means over six benchmarks and three seeds (TEST split). The sleeper is omitted because a history-fitted oracle is invalid by construction; see Table 3.

RACE-D never differs significantly from RACE (0 non-ties in 252 cells). Against camouflage it helps at $f = 0.7$ (+5.1 points on average) but hurts at $f = 0.3$ (-3.7). Overall it wins 7 and loses 17 of 36 camouflage cells. It also hurts against echo (-5.5 at $f = 0.7$), and against sleepers at $f = 0.3$ (-14.2). Across all 324 cells it wins 14 and loses 31. Restricting to disagreement removes the questions that pin down the honest peers’ accuracies, trading one misspecification for another. We keep the frozen RACE as the method. RACE-D was specified and implemented before E9 was run, and RACE was not modified afterwards (Appendix A).

7.8 E11: classical crowdsourcing estimators

Is the receiver anchor necessary, or would any mature crowdsourcing estimator do? E11 adds four classical label-free estimators to both Dawid–Skene variants: iterative weighted majority voting (IWMV) [27], MACE [17], GLAD [38] and, on binary questions, the message-passing estimator of Karger, Oh and Shah (KOS) [21]. The protocol is exactly that of RACE. Each honest agent runs every estimator on its own HISTORY window of peer answers, no estimator sees a label, and ties go to the agent’s own answer. E11 covers 342 worlds: six benchmarks; a coherent bloc at $f \in \{0.1, 0.3, 0.5, 0.7\}$; gate-aware liars ($p = 0.25$) and the four LLM prompts at $f \in \{0.3, 0.5, 0.7\}$; and three seeds. Figure 12 shows the result.

E11. Classical crowdsourcing estimators collapse when liars are the majority; the anchor does not



Means over six benchmarks, six attack settings (coherent, gate-aware $p = 0.25$, four LLM prompts) and three seeds, TEST split. KOS is defined for binary questions only; full-confusion Dawid-Skene is not run on open-answer benchmarks (n/a).

Figure 12: E11. Classical crowdsourcing estimators against RACE. (a) Accuracy of an honest agent as the liar share f grows. The yellow band spans IWMV, MACE, GLAD and label-free Dawid-Skene; the dashed line is the best of them. (b) Accuracy per benchmark at $f = 0.7$. KOS is defined for binary questions only, and full-confusion Dawid-Skene is not run on open-answer benchmarks. Means over attacks and seeds, TEST split.

With a liar minority the classical estimators are as good as Race. At $f = 0.3$ the best classical estimator scores 92.3% and RACE 92.9%. Over the 203 comparisons between RACE and a classical estimator at $f \leq 0.3$, RACE ties in 192 and wins 11. This is the regime these estimators were designed for, and they work.

With a liar majority every one of them collapses. At $f = 0.7$ the best classical estimator scores 17.6% (majority vote 22.8%, full-confusion Dawid-Skene 23.6% on the categorical benchmarks). RACE scores 89.5%, the known-channel oracle 90.7% and the receiver alone 84.2%. On BoolQ, KOS scores 25.3% against RACE’s 91.7%. The failure is the label-switching symmetry of Proposition 3, not a weakness of any single estimator. The likelihood of a latent-class model is invariant to relabelling the truth. Without an anchor, EM settles on the relabelling in which the *majority* is reliable, so it decodes the liars’ labelling faithfully and inverts every honest agent. The receiver anchor breaks exactly this symmetry. Anchoring Dawid-Skene and nothing else gives RACE without clone tempering, which E1–E3 already evaluate.

Paired tests. There are 551 (benchmark, f , attack) comparisons between RACE and a classical estimator. RACE wins 255 and loses 0, with Holm correction over all 893 comparisons of the study. At $f \geq 0.5$ it wins 244 of 348. The classical estimators are therefore not stronger baselines than those of E1–E3. Their failure at $f \geq 1/2$ is the reason an anchored estimator is needed.

7.9 E6: changing attackers

The sleeper is the one attack in the zoo that defeats every static defender, including the known-channel oracle, because it violates stationarity. We therefore re-ran five attacker schedules under a strictly causal protocol on MMLU, MedQA, BoolQ and MATH-500 ($f \in \{0.3, 0.5, 0.7\}$, two seeds; 120 streams of 200 questions). At each step every honest receiver predicts from a state fitted only on

Non-stationary attackers ($f = 0.5$, predict-then-commit)

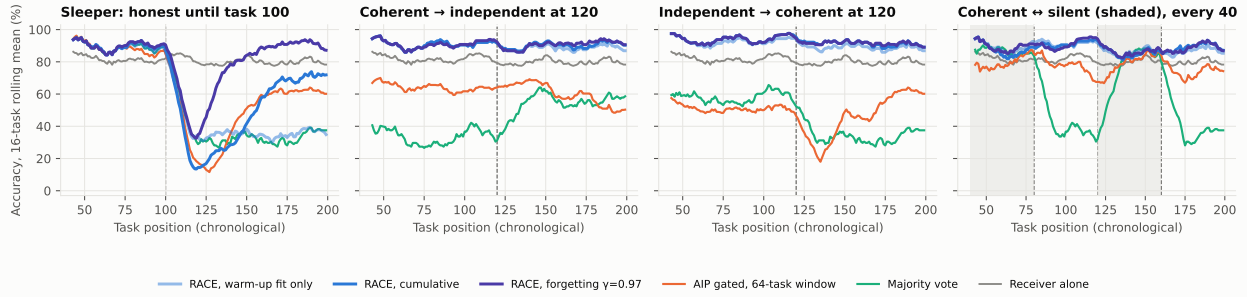


Figure 13: E6. Causal online accuracy (16-question rolling mean) at $f = 0.5$ under four non-stationary attacker schedules, averaged over four benchmarks and two seeds. Dashed lines: the attacker’s switches. Shaded (right): the bloc is silent.

earlier questions, then commits the observation. Methods refit every 8 questions after a 40-question warm-up. A test confirms that future broadcasts or labels cannot change a past prediction.

Sleepers. The sleeper answers honestly for 100 questions and then lies coherently. We average over the last 60 questions, which start 40 questions after the switch, and over the three values of f . A RACE fit frozen at warm-up scores 45.2%. Cumulative refitting recovers to 70.8%. Forgetting ($\gamma = 0.97$) recovers to 88.3% and a 64-question window to 86.1%, both above the receiver alone (79.8%) and far above a windowed AIP (69.4%) and majority vote (43.6%). At $f = 0.7$ the frozen fit and majority vote both score 0.0%, while forgetting scores 86.5% (receiver alone: 78.9%). Once the sleeper starts lying, it is a coherent bloc like any other, and the forgetting receiver learns to read it backwards within about 40 questions (Figure 13).

Regime switches and the price of forgetting. When the bloc switches from coherent to independent lies, or back, forgetting scores 90.2% and 89.9% after the switch (cumulative: 89.6% and 89.2%; windowed AIP: 64.7% and 58.1%). A bloc that alternates between coherent lies and silence every 40 questions leaves RACE at 85.9%, against 62.5% for majority vote. On a stationary coherent schedule, forgetting costs nothing measurable (88.8% vs. 88.4% cumulative). Forgetting is therefore the default we recommend when stationarity cannot be assumed, and it is exposed as `TrustLayer(forgetting=0.97)`.

8 E8: a live multi-agent LLM swarm

The replay studies reuse cached answers from the original artifact. E8 generates new ones. Six small instruction-tuned models from six labs sit on one panel: Qwen2.5-1.5B, SmolLM2-1.7B, Granite-3.3-2B, OLMo-2-1B, Llama-3.2-1B and Gemma-3-1B. They answer the first 240 questions of MMLU and BoolQ (120 each) on a CPU, in float32. Every answer is the arg-max of the next-token distribution over the option letters. The question prefix is encoded once, and the key-value cache is branched for each role’s instruction, so every role sees exactly the same question. The run started after RACE was frozen, and nothing about it was inspected before the evaluation below.

Roles. In round 1 each model answers twice. As an *honest* agent it answers the question. As a *solo* saboteur it is told to work out the answer privately and give a plausible wrong option. In

Table 2: Significant wins/ties/losses of RACE against each baseline (Holm-corrected Wilcoxon signed-rank and paired bootstrap interval must agree, $\alpha = 0.05$, task as unit).

Study	verdict Baseline	Wins	Ties	Losses	Cells
E3 LLM deceivers	AIP gated	37	59	0	96
	AIP soft gate (M1)	28	68	0	96
	Dawid-Skene (label-free)	35	61	0	96
	Majority vote	38	58	0	96
	Receiver alone	59	37	0	96
E1 coherent sweep	AIP gated	20	38	2	60
	AIP soft gate (M1)	14	44	2	60
	Dawid-Skene (label-free)	31	26	3	60
	Majority vote	41	18	1	60
	Receiver alone	37	23	0	60
E2 attack zoo	AIP gated	100	148	4	252
	AIP soft gate (M1)	73	176	3	252
	Dawid-Skene (label-free)	87	157	8	252
	Majority vote	121	129	2	252
	Receiver alone	136	96	20	252

round 2 each model answers twice more. As a *rushing* saboteur it first sees the honest panel’s votes and picks the wrong option most likely to mislead them. As a *debating* honest agent it first sees all round-1 votes, half of them from solo saboteurs, and is told only that some panelists may be unreliable. A *colluding* bloc needs no further inference: every colluder adopts the coalition leader’s solo lie.

Worlds. The live answers are evaluated with the same harness as the replay studies. Ten identities are drawn from the six-model roster. For independent answers we cross the Byzantine fraction $f \in \{0.1, 0.3, 0.5, 0.7\}$ with four attacks: solo, rushing, colluding LLM saboteurs, and the symbolic coherent bloc. For post-debate answers we use $f = 0.5$ with solo and colluding saboteurs. Each setting has five seeds. The HISTORY/VALIDATION/TEST split and all statistics are as in Section 6.

What the agents did. Honest accuracy is low and spread widely: 32.5–60.8% on MMLU and 49.2–85.8% on BoolQ. Instructed saboteurs lie only partly (Figure 14, top). On MMLU, solo and rushing saboteurs are right 29.0% and 24.9% of the time, against 25% at chance, and 50% of the saboteur channels are below chance. On binary BoolQ *none* are: saboteurs are right 58.6% (solo) and 57.4% (rushing) of the time. The models do not so much lie as fall back on a fixed option. Granite-3.3-2B as a saboteur says “yes” on 1.7% of BoolQ questions. SmolLM2-1.7B as an *honest* agent says “yes” on 89.2% of them (the gold share is 45.8%): it is right on 96.4% of yes-questions and 16.9% of no-questions. Channels like these carry information through their asymmetry, which the symmetric one-coin model of Section 4 cannot represent. In the debate round, honest agents changed their answer on 16.0% of questions, but only 3.7% moved to the saboteurs’ plurality answer. Their accuracy barely changed (56.9% before, 55.7% after).

MMLU: the thesis holds with weak agents. With independent answers and LLM saboteurs at $f = 0.5$, RACE scores 54.2%. That is above the receiver alone (43.4%), majority vote (46.0%),

AIP (41.8%) and label-free Dawid–Skene (44.9%). It matches the known-channel oracle (53.9%). At $f = 0.7$ the figures are 50.4%, 43.8%, 40.6%, 40.8% and 30.3%. Averaged over LLM saboteurs at $f \in \{0.3, 0.5, 0.7\}$, every honest receiver gains, by 5.6 to 27.9 points. Against the symbolic coherent bloc at $f = 0.5$, majority vote falls to 8.3% while RACE scores 56.4%.

BoolQ: a misspecification the live data exposed. Against the coherent bloc, RACE reads the lie backwards as in the replay studies: 84.8% at $f = 0.5$, against 60.1% alone and 20.0% for majority vote. Against LLM saboteurs, however, the frozen v3.0 fell *below* the receiver alone: 56.2% against 60.1% at $f = 0.5$, while majority vote reached 69.3%. The strongest receiver lost most (Granite-3.3-2B: 83.3% alone, 68.3% with v3.0), and the weakest gained, so this is not the weak-anchor failure of Section 9. The cause is the option bias above. A channel that says “yes” regardless of the truth has one-coin accuracy close to $1/2$, so its information sits in its asymmetry, which the one-coin model discards. Class-conditional channels model that asymmetry. With them (v3.1, Section 4.3), RACE scores 69.5% at $f = 0.5$ and 65.8% at $f = 0.7$ (majority: 62.4%), and the strongest receiver scores 74.5%. We adopted this rule after seeing these data, so this is not a confirmation of v3.1. On replayed BoolQ, v3.1 is never significantly worse than v3.0: 1 of 78 cells favour v3.1 and 0 favour v3.0 (Appendix C). Each world has only 48 test questions, so few per-cell differences are significant. Across the 32 cells per baseline, RACE (v3.1) loses 0 to the receiver alone, 0 to majority vote, 0 to AIP and 0 to label-free Dawid–Skene. It wins 5, 7, 7 and 6 cells respectively.

Debate: it depends on whether the saboteurs carry signal. We compare post-debate with independent answers on the matched attacks (solo and colluding saboteurs, $f = 0.5$). On MMLU, debate lowers RACE from 56.0% to 49.6%, majority vote from 48.3% to 45.4% and the receiver alone from 43.4% to 41.6%. There the saboteurs’ answers are informative (half of them below chance), and after debate the honest answers are no longer independent draws, so a receiver-local estimator has less to work with. On BoolQ the saboteurs carry almost no signal, and debate *helps*: RACE rises from 69.0% to 73.2% and majority vote from 70.4% to 73.1%. The safe protocol is to record each agent’s independent answer before any discussion. Pooling can then use the independent answers, the post-debate answers, or both.

9 Where RACE fails, and other limitations

Theorem 1 leaves the adversary three levers. We found each one working against RACE.

Violating conditional independence: camouflage. The camouflage attacker tells the truth whenever the honest vote is clear, earning a high estimated accuracy. When the honest vote is contested, it coherently backs the strongest wrong honest answer. Its reports depend on the honest reports, not only on the truth, so the one-coin model misreads it as an expert. At $f = 0.7$ it pushes RACE 6.3 points below the receiver alone. Capping every peer’s accuracy at the receiver’s own (RACE+self-cap) recovers part of the loss. No deployable method we tested beats the receiver alone under this attack at $f = 0.7$. The obvious repair, fitting channels only on questions with disagreement (RACE-D, Section 7.7), helps at $f = 0.7$ but hurts at lower f and against echo attackers. A channel model conditioned on the honest vote pattern, rather than a filter on questions, remains open.

E8: a live six-model swarm (fresh CPU inference). RACE v3.0 was frozen before this run; v3.1's binary rule was adopted after it

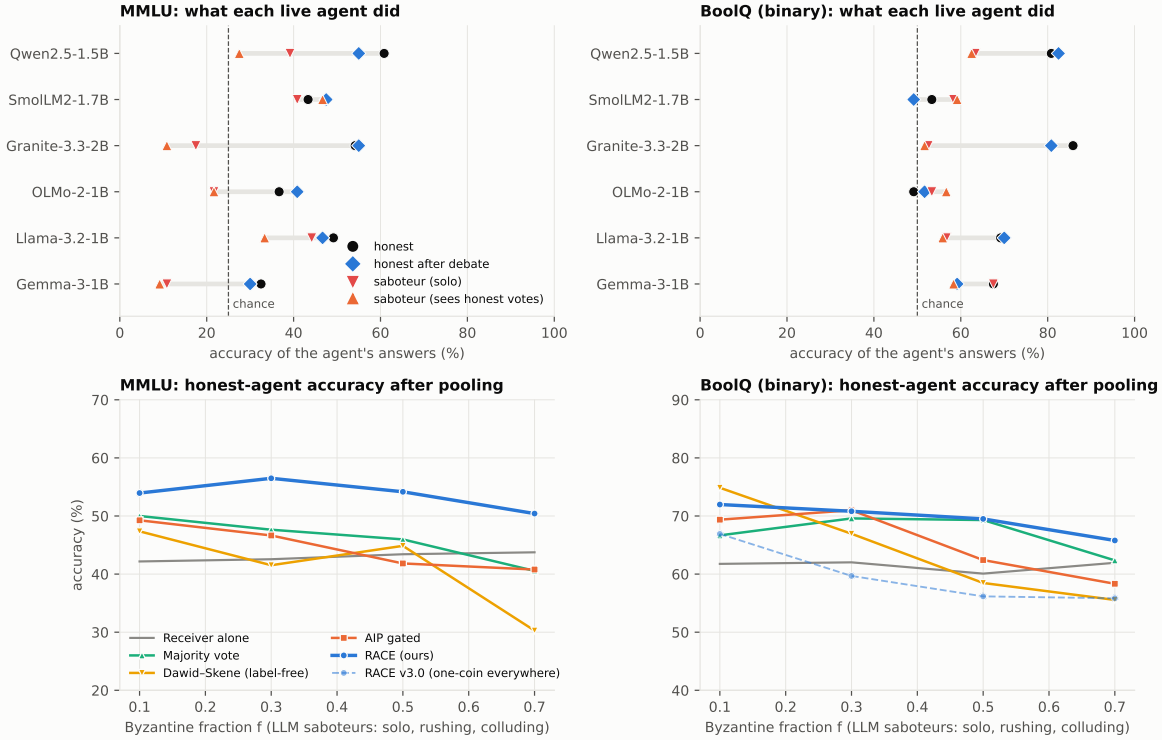


Figure 14: E8. Live six-model swarm. Top: accuracy of each model’s answers in each role; saboteurs to the left of the chance line lie reliably. Bottom: honest-agent accuracy after pooling, against LLM saboteurs (mean over solo, rushing and colluding), 48 test questions per world, five seeds. Dashed: RACE v3.0, frozen before this run; on MMLU it coincides with v3.1. The binary rule of v3.1 was adopted after this run.

Violating stationarity: sleepers. A sleeper that behaves honestly on history and lies afterwards defeats every static fit, including the known-channel oracle, which is fitted on the same history (Figure 3). The causal online study (Section 7.9) shows that refitting with forgetting recovers, with a lag set by the forgetting rate.

Weak anchors. Identification requires the receiver to be better than chance on the *effective* answer space (Proposition 3). On open-answer questions the candidate set is often just {consensus, one alternative}. There a receiver near 50% is effectively at chance. A rushing attacker that lies with that receiver’s own wrong answers then produces a labelling that is observationally equivalent to the truth from that receiver’s viewpoint. In our MATH-500 runs this affects the weakest roster member (Llama-3.2-3B, 44.5% on MATH-500). Stronger receivers in the same swarm are unaffected. Figure 8 quantifies the gain as a function of receiver competence. It is also why RACE trails AIP and majority on MATH-500 at small f .

Option-biased channels, and a post-hoc amendment. The one-coin model gives each channel one accuracy and spreads its errors evenly over the alternatives. The live swarm (Section 8) showed that small models violate this on binary questions: some answer “yes” or “no” largely regardless of the truth. There the frozen v3.0 fell below the receiver alone (56.2% vs. 60.1% at $f = 0.5$). We amended RACE to use class-conditional channels on binary questions (v3.1, Sec-

tion 4.3), which scores 69.5% there. On the 78 replayed BoolQ cells it is never significantly worse than v3.0 (1 significant gains, 0 losses). The amendment was chosen after seeing the live data, so the live BoolQ result is not a confirmation of v3.1; a fresh confirmation set is needed. Even with v3.1, the strongest live BoolQ receiver loses by pooling with a near-chance crowd (83.3% alone, 74.5% with RACE).

Scope. All replay studies use cached single-turn answers. Receivers aggregate final answers, not reasoning, and the symbolic attackers are gold-aware by design. The live study (Section 8) uses small 1–2B models on a CPU and 120 questions per benchmark, so it is a confirmation, not a benchmark. Open-answer scoring relies on an approximate symbolic equivalence checker. Correlated honest errors (shared attractors) remain partly unmodelled: clone tempering only removes near-duplicates. The known-channel oracle shows the remaining headroom.

10 Audit of the prior artifact

A journal-grade result on cached model outputs is only as sound as the cache and the code that reads it. Before building on the prior releases [8, 30] we audited them. Every item below was verified by running code or diffing files.

Truncation (X3). In the original cache, 944 of 13,500 answers were *manufactured*: a 640/2048-token budget truncated completions, and the extractor then read a number out of the unfinished working, at a 100% parse rate. The upstream repair regenerated 2,313 rows. Our loader reproduces the repaired single-model table exactly. All our results use the repaired cache. MATH-500 conclusions in the earlier derivative release, which predates the repair, are superseded. For example, three strong models gained 26–50 points on MATH-500.

Incomplete recompute. After the repair, four result files that the manuscript reads (`n_scaling`, `minimax_regret`, `accuracy_landscape`, `gate_aware_mitigations`) are byte-identical to their pre-repair versions. All 54,432 frozen-roster rows of the aggregation sweep are value-identical, because the sweep resumed and skipped existing cells. The headline macros of the upstream manuscript are therefore pre-repair. A fresh recompute of the gate-aware anchor gives 0.087/0.889 where the paper prints 0.082/0.878. We re-derive every number from scratch.

Code defects. With $p_{\text{obs}} < 1$ the upstream AIP fit paired answers from *different* questions. It seeded the randomised threshold with Python’s salted `hash()`, so runs were not reproducible across processes. It kept stale state across refits, and it counted missing answers as disagreements. The derivative release fixed all four. We build on the fixed code, and all our seeds are SHA-256-derived.

Nondeterminism. Regenerated Llama-3.2-3B rows flip their extracted answer in 52 of 300 re-runs (17.3%) at identical settings. We report this and treat the regenerated rows as fresh samples.

Pipeline extension (MX). Same-model “replicas” were byte-identical because one seed was shared across the batch, so the negative controls passed trivially. The extension stopped before any liar ran. Our live experiment uses distinct models and explicit roles instead.

None of these defects changes the qualitative picture of AIP (inversion helps against a coherent bloc; the gate-aware attacker defeats it). All of them change numbers. The repaired, re-derived numbers are the ones reported here.

11 Conclusion

“Liars are information” is true in a precise sense. A lie is information exactly to the extent that it depends on the truth, and a receiver that knows it is honest can measure that dependence

without labels. Replacing a coherence gate with receiver-anchored channel estimation removes the binary collapse and the gate-aware defeat of the prior mechanism. Across all six benchmarks and Byzantine fractions up to 0.9, it is never significantly worse than a receiver that trusts only itself, with three exceptions: the camouflage and sleeper attacks, and one open-answer cell where the receiver cannot anchor. It also holds up on real LLM deceivers that were not inspected during development. The information budget makes the title measurable. Against independent liars at $f = 0.7$, RACE does better than an oracle that knows who lies and removes them, which no amount of filtering can achieve in expectation. The same analysis says where the idea must stop. An attacker that answers independently of the truth can only silence itself. An attacker whose lies depend on the honest votes, or change over time, attacks the estimator’s assumptions rather than its weights. A receiver at chance cannot anchor anything. The live swarm adds another: small models answer binary questions with a fixed option bias, which a symmetric channel model cannot represent. Three directions follow. First, camouflage calls for channel models conditioned on the honest vote pattern; restricting the fit to contested questions (RACE-D) is not enough. Second, binary questions call for class-conditional channels, which recovered the live BoolQ loss in our ablation. Third, protocols should record independent answers before agents see each other: on MMLU, debate removed information that pooling needed (Section 8).

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A Development log: changes made after inspecting data

A change was admissible only if all three of the following held:

1. it fixes a mechanism that can be stated without reference to a benchmark number;
 2. it keeps every previously fixed hyper-parameter;
 3. the superseded variant stays in every table.
1. **Initial design.** Anchored one-coin EM with clone tempering by raw agreement ≥ 0.95 (kept as `race_rawclone`).
 2. **Rejected: multi-start MAP.** We ran EM from both the receiver’s answers and the plurality and kept the higher log-posterior (`race_ms`). Trigger: a near-chance MATH-500 receiver captured by a rushing liar. It did not help, because with a near-chance anchor the two labellings are likelihood-equivalent. It was harmful when replicated channels inflate the likelihood of the adversarial labelling. The anchor therefore acts through initialisation and constraint, not through a likelihood comparison.
 3. **Adopted: two-part clone rule.** Same-source links (raw agreement ≥ 0.99 , applied before any fit) and dependent-error links (agreement ≥ 0.95 on tasks where a member errs). Trigger: GSM8K with uninformative/partial liars at $f = 0.7$, where raw-agreement grouping merged distinct accurate models and collapsed honest evidence (Proposition 7).

The method was then frozen, and every study was re-run from scratch. One extension was added after the freeze, as a separate method that leaves RACE unchanged: RACE-D (Section 7.7), which fits channels on disagreement questions only. It was specified before E9 was run and is reported as a negative result. A second post-freeze change is the binary rule of Section 4.3 (v3.1), adopted after the live swarm (E8) exposed option-biased channels; every study was re-run with it, and v3.0 is reported alongside (Appendix C). The replayed LLM deceivers (E3) and the live swarm (E8) had not been inspected at freeze time. On E3 the adopted rule outperforms both superseded variants at every $f \geq 0.5$.

B Information budget by attack

Table 3: E9 by attack setting (means over six benchmarks and three seeds, TEST). B : information budget, the known-channel oracle minus the liar-removal oracle. Δ : RACE minus the liar-removal oracle. Liar-removal oracle: 93.5%, 92.6% and 89.7% at $f = 0.3, 0.5, 0.7$. Positive Δ : RACE does better than removing every liar. The two camouflage settings coincide at $f = 0.7$: with three honest agents the plurality share is 1/3, 2/3 or 1, so both thresholds trigger on the same questions.

Attack	$f = 0.3$		$f = 0.5$		$f = 0.7$	
	B	Δ	B	Δ	B	Δ
Independent wrong answers	+1.8	+1.1	+2.6	+1.9	+5.6	+5.1
Gate-aware, $p = 0$	+1.7	+1.1	+2.4	+2.2	+5.8	+5.4
Gate-aware, $p = 0.5$	+1.7	+1.2	+2.5	+2.1	+4.3	+3.8
Gate-aware, $p = 1$ (coherent)	+0.9	-1.2	+1.0	-0.9	+2.2	+1.2
Attractor	+1.8	-1.0	+2.2	-0.2	+3.9	+1.8
Partial liar, $e = 0.8$	+0.2	-2.1	+0.5	-2.1	+1.2	-0.4
Partial liar, $e = 0.5$	+0.0	-1.5	-0.4	-1.5	+0.7	-2.4
Uninformative (silent)	-0.2	-2.8	-0.7	-2.2	-1.6	-4.7
LLM: always wrong	-0.1	-1.1	-0.3	-1.4	-0.9	-2.1
LLM: rushing	-0.1	-0.8	-1.3	-1.7	-0.4	-1.5
LLM: hallucination	-0.1	-0.7	-0.2	-0.9	+0.1	-1.6
LLM: negation	-0.2	-0.9	-0.3	-0.7	+0.1	-1.4
Echo (copies an honest agent)	-2.2	-2.7	-2.8	-3.2	-1.8	-2.3
Camouflage, $\theta = 0.9$	-1.0	-2.9	-2.0	-5.6	-4.1	-13.0
Camouflage, $\theta = 0.7$	-2.2	-4.0	-4.8	-6.7	-4.1	-13.0
Sleeper (non-stationary)	-3.3	-4.2	-34.1	-53.3	-89.7	-89.7

C RACE v3.1 versus v3.0 on binary questions

Table 4: Mean BoolQ accuracy (% , TEST) of the receiver alone, the frozen v3.0 (one-coin channels everywhere) and v3.1 (class-conditional channels on binary questions). On the other benchmarks the two versions coincide by construction. Replayed BoolQ cells of E1–E3 with a significant v3.1–v3.0 difference: 1 of 78.

Study	Receiver alone	v3.0	v3.1	Change
E1 coherent sweep	88.8	93.7	93.7	+0.0
E2 attack zoo	89.1	88.2	88.9	+0.6
E3 LLM deceivers	89.1	91.7	91.9	+0.2
E9 information budget	89.1	88.9	89.5	+0.6
E8 live swarm (LLM saboteurs)	61.5	59.6	69.5	+9.9

D Hyper-parameters and implementation

Anchor initial posterior on own answer	0.75
Anchor prior (receiver accuracy)	Beta, mean 0.75, strength 8
Anchor constraint	$a_i \geq 1/K + 0.05$
Peer prior	Beta, mean = chance, strength 2
Same-source / dependent-error thresholds	0.99 / 0.95
Minimum support for a link	8 tasks
Accuracy clip	$[10^{-3}, 1 - 10^{-3}]$
EM	at most 200 iterations, tolerance 10^{-6}
Decision band for reporting TRUST/INVERT	$ \lambda > 0.25$ (diagnostic only)
Online forgetting (E6)	$\gamma = 0.97$, refit every 8 tasks

E Reproducibility

All code, data, results, figures, videos and this manuscript are in the accompanying repository and Hugging Face release. One command, `experiments/reproduce_all.sh`, regenerates everything on a 4-core CPU without a GPU in about an hour: E1–E7 and E9 from the cached answers, the E8 evaluation from the live-swarm cache, every figure, table and number, the videos, the repository README and this PDF. It finishes by re-checking every qualitative claim in the text against the results (`experiments/verify_claims.py`). `experiments/live_swarm.py` regenerates the live answers themselves. The numbers in this paper are macros written by `experiments/make_numbers.py`. Each study directory holds a `run_manifest.json` with the input and code hashes, wall time and peak memory. The repository’s test suite covers RACE’s identification, invariance, clone-grouping, class-conditional and causality properties.

F Compute resources

Everything ran on one 4-core x86-64 CPU with 15 GB of memory and no GPU. The replayed studies (E1–E7, E9, E11 and the E6 streams) took 47 minutes of wall-clock time in total, as recorded in each study’s run manifest. The live swarm E8 took 3.1 CPU hours of inference for six models in float32. The per-study times, code hashes and input hashes are listed in the repository’s evidence ledger (`docs/EVIDENCE.md`).

G Reproducibility checklist

- **Code and data.** All code, cached model answers, live-swarm answers, per-task results, figures and videos are released under MIT; benchmark items keep their upstream licenses.
- **One command.** `experiments/reproduce_all.sh` regenerates every study, number, table, figure, video, the README and this PDF, then re-checks every qualitative claim.
- **Randomness.** All seeds derive from SHA-256 of the world specification; runs are bit-reproducible across processes.
- **Splits.** HISTORY/VALIDATION/TEST are fixed by a hash of the question identifier. Methods never see labels; attacks are chosen on VALIDATION; everything is scored on TEST.
- **Hyper-parameters.** Fixed before evaluation and never tuned (Appendix D). Every post-hoc change is logged with its trigger (Appendix A), and superseded variants are reported.

- **Statistics.** Paired bootstrap intervals (2,000 resamples) and Wilcoxon signed-rank tests, Holm-corrected within each study. A verdict requires both to agree, and the unit is the question.
- **Confirmation.** Two confirmation sets were not inspected during the design of v3.0. A third (E10) was pre-registered before any of its answers existed.
- **Claims.** Every number in the text is a generated macro, and every qualitative sentence is re-derived by `experiments/verify_claims.py`.
- **Tests.** The test suite covers identification, invariance, clone grouping, the binary rule, the classical baselines, causality of the online protocol and the evidence decomposition.

H Ethics and broader impact

The deceptive answers we study come from language models that were explicitly prompted to mislead. They are used to evaluate defences, and no deceptive content was shown to people. No human subjects or personal data are involved. The work has a dual use. The information budget and the camouflage attack show an adversary what RACE can and cannot see, and the same knowledge that makes the defence auditable helps an attacker design around it. We report the failure modes prominently for this reason. A deployed system should combine receiver-local pooling with the online forgetting variant and with independent-answer protocols, and it should treat a negative information budget as an alarm. RACE decides from agreement patterns alone. It is a statistical defence and not a guarantee, and it should not be the only safeguard in high-stakes settings.